



the title of my thesis project which may
span multiple lines

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Chapter 1

Introduction

1.1 Viral Vectors in Emerging Therapies

Clinical medicine progressively transitioning from broad, non-specific treatments such as chemotherapy or immunosuppressants to more refined, high-fidelity solutions. Those include small molecules targeting pathways of the immune system, antibody treatments, and more recently, gene therapy and virotherapy. Virotherapy emerges as a potentially promising approach for treating various cancers [1], while gene therapy shows promise for hereditary diseases [2], and genetic vaccination is being explored for the treatment and prevention of infectious diseases [3]. Gene therapy utilizes a vector to deliver genetic material to the target cell. Numerous vector systems are currently in use and are broadly categorized into viral and non-viral vectors. Non-viral vectors are further categorized into physical and chemical vectors. Viral vectors are particularly favored due to their evolutionary adaptation for efficient cell transduction as intracellular parasites [4]. The most prevalent viral vectors employed in clinical trials are adenoviral vectors, followed by adeno-associated virus and lentiviral vectors [5].

1.2 Adenoviruses as Vectors

1.3 Producer Cell Lines

Test commit

1.4 CRISPR sequences and Cas proteins

In brief, the CRISPR-Cas-System can be described as either of two things: originally, it is a component of the bacterial arsenal to defend against foreign genetic material. After its discovery, it has been studied and employed to accurately and precisely induce strand breaks in DNA, as a means of facilitating genetic engineering. its components are the growing family of Cas (CRISPR-associated) nucleases which do the actual strand-nicking and the CRISPR-Sequences, "CRISPR" meaning *Clustered Regularly Interspaced Short palindromic Repeats*. Long before the role of this system was understood, the repetitive interspaced sequences were discovered in *Escherichia coli* [6]. The discovery of similar sequences in archaea sparked new interest in the topic, as the presence of similar sequences in phylogenetically distant organisms implies a high level of evolutionary conservation and as such a high biological importance [7]. It was later described, that the highly varied spacer sequences represent genetic information of foreign (viral) origin [8] [9] and are accompanied by various associated (Cas) genes [10] that execute their function either in maintaining and expanding the CRISPR loci, called class I, or serving as direct effector molecules in the cleavage of foreign DNA or RNA - class II [11]. At present time, over 30 different Cas genes of both classes and a multitude of subtypes have been described [12]. other essential components of the system are the CRISPR-RNA (crRNA) and the trans-activating crRNA (tracrRNA). the crRNA is complimentary in sequence to the targeted foreign DNA or RNA (the protospacer), is derived from the spacer region of the CRISPR locus and has a length of $\simeq 20$ bp. It directs the effector complex to its target [13]. The tracrRNA is important for the maturation of the crRNA and assembly of the final functional effector ribnucleoproteincomplex of Cas protein and associated RNAs [14]. additionally, the protospacer must be neighbouring an adjacent motif (PAM) varying in sequence depending on the Cas type and class. The PAM-motifs are not integrated into the spacer sequences but are recognized by the Cas proteins. For example the Cas9 of *s. pyogenes* requires a NGG PAM sequence directly downstream of the crRNA binding site and cleaves 3 base pairs upstream of the PAM. Since crRNA and tracrRNA are two separate molecules, it was then proposed to combine the two into a single chimeric guide RNA (sgRNA) to facilitate easier application of the system *in vitro* [15].

To apply the system to edit eukaryotic DNA, the Cas9 sequence is codon optimized and paired with a nuclear localization signal [16]. For our application, genetically modifying HEK293 cells, we chose Cas9 as it targets DNA and is the most popular and well studied Cas system to date.

1.5 The hit and run hypothesis

This hypothesis is in contrast to the general consensus that persistent presence of viral gene products is necessary for maintaining a transformative cellular phenotype. It has first been proposed by Skinner et al. [17] in the context of HSV-2 mediated transformation of primary hamster fibroblasts. In the realm of adenovirus, ...

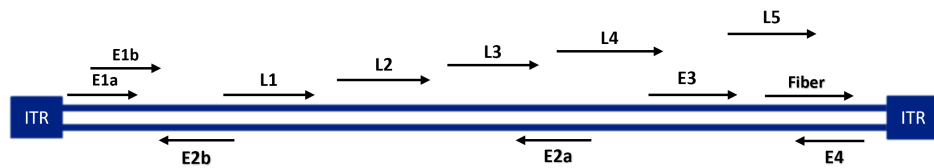


Figure 1.1: Schematic representation of the genome of adenovirus type 5

Chapter 2

Materials and Methods

2.1 Materials

2.1.1 Reagents

Table 2.1: Reagents used in this Project

Reagent	Manufacturer
Acetic acid 100%	AppliChem, Darmstadt, Germany
Acrylamid	???
Agar	AppliChem, Darmstadt, Germany
Agarose LE	Genaxxon, Ulm, Germany
Ammonium persulfate	Merck, Darmstadt, Germany
Ampuwa	Fresenius, Bad Homburg v.d.H., Germany
APS	???
BSA	
Bromphenol blue	Merck, Darmstadt, Germany
Chloroform	AppliChem, Darmstadt, Germany
EDTA disodium dihydrate	AppliChem, Darmstadt, Germany
Ethanol 100%	AppliChem, Darmstadt, Germany
Ethidium bromide 0.07%	AppliChem, Darmstadt, Germany
Liquid nitrogen	Air Products, Allentown, USA
Formaldehyde 37%	AppliChem, Darmstadt, Germany
Glucose	AppliChem, Darmstadt, Germany

Table 2.1: Continuation

Reagent	Manufacturer
Glycerin 87%	AppliChem, Darmstadt, Germany
Glycerin, H ₂ O free	AppliChem, Darmstadt, Germany
HEPES	AppliChem, Darmstadt, Germany
Isoamyl alcohol	Riedel-de Haen, Seelze, Germany
Isopropyl alcohol	AppliChem, Darmstadt, German
L-arabinose	Sigma-Aldrich, St. Louis, USA
LB Agar	Invitrogen, Waltham, USA
Lysozyme	Sigma-Aldrich, St. Louis, US
Methanol	Fisher Scientific, Hampton, USA
Paraformaldehyd	
Polyacrylamide	
Phenol	AppliChem, Darmstadt, Germany
Polybrene	Sigma-Aldrich, St. Louis, USA
Potassium acetate	AppliChem, Darmstadt, Germany
Potassium chlorate	AppliChem, Darmstadt, Germany
SDS ultrapure	AppliChem, Darmstadt, Germany
Silver nitrate	AppliChem, Darmstadt, Germany
Sodium acetate	AppliChem, Darmstadt, Germany
Sodium carbonate	AppliChem, Darmstadt, Germany
Sodium chloride	AppliChem, Darmstadt, Germany
Sodium hydroxide	AppliChem, Darmstadt, Germany
Sodium thiosulfate	AppliChem, Darmstadt, Germany
TEMED	AppliChem, Darmstadt, Germany
Tris ultrapure	AppliChem, Darmstadt, Germany
Triton X-100	
Tween 20	AppliChem, Darmstadt, Germany
Xylencyanol	AppliChem, Darmstadt, Germany

2.1.2 Enzymes and Buffers

Table 2.2: Enzymes and Enzyme Buffers used in this Project

Product	Supplier
10x buffer R	Thermo Scientific, Rockford, USA
10x r2.1 buffer	New England BioLabs, Ipswich, USA
BbsI restriction enzyme	New England BioLabs, Ipswich, USA
GoTaq buffer	Promega, Mannheim, Germany
GoTaq DNA Polymerase	Promega, Mannheim, Germany
PstI restriction enzyme	Thermo Scientific, Rockford, USA
Proteinase K	AppliChem, Darmstadt, Germany
Q5 buffer	New England BioLabs, Ipswich, USA
Q5 DNA Polymerase	New England BioLabs, Ipswich, USA
Ribonuklease A (RNAse)	AppliChem, Darmstadt, Germany
T4 ligase	
10x T4 buffer	

2.1.3 Antibiotics

Table 2.3: Antibiotics used in this Project

Antibiotic	Supplier
Puromycin	
Streptomycin	Roth, Karlsruhe, Germany

2.1.4 Appliances

Table 2.4: Appliances used in this Project

Appliance	Manufacturer
Amersham nitrocellulose membrane 0.45 μ m	??
Avanti J-25 Centrifuge	Beckmann Coulter, Brea, USA

Table 2.4: Continuation

Appliance	Manufacturer
Biological safety cabinet Hera Safe	Heraeus, Hanau, Germany
Biophotometer	Eppendorf, Hamburg, Germany
Centrifuge 5415 R	Eppendorf, Hamburg, Germany
Cytoflex FACS	Beckmann Coulter, Brea, USA
Electrophoresis Power Supply	ThermoScientific, Rockford, USA
Eporator	Eppendorf, Hamburg, Germany
Gel Doc EZ Imager	BioRad, Hercules, USA
Gel electrophoresis large gel box	BioStep, Burkhardtsdorf Germany
Gel electrophoresis small gel box	BioStep, Burkhardtsdorf Germany
Heraeus Incubator	Thermo Scientific, Rockford, USA
Herafreeze Basic -80	Thermo Scientific, Rockford, USA
Incubator Shaker G25	NBS, Edison, USA
Inolab pH-Meter	WTW, Weilheim, Germany
Laboratory scale Sartorius	Sartorius, Göttingen, Germany
Leitz DMIL fluorescence microscope	Leica, Wetzlar, Germany
Leitz DMIL light microscope	Leica, Wetzlar, Germany
Magnet and magnetic stirrer RCT IKAMAG	IKA, Staufen, Germany
Megafone 40 Centrifuge	Heraeus, Hanau, Germany
Mikro 220R Centrifuge	Eppendorf, Hamburg Germany
Milli-Q direct	Merck, Darmstadt, Germany
Mixer HC thermomixer	StarLab, Brussels, Belgium
improved Neubauer chamber	???
Odyssey CLx Imager	LI-COR Biosciences, Lincoln, USA
Optima XPN-80 Ultracentrifuge	Beckman Coulter, Brea, USA
PD-10 columns	GE Healthcare, Chicago, USA
PowerPac 200 Electrophoresis Power Supply	BioRad, Hercules, USA
RC 5C Plus Centrifuge	ThermoScientific Rockford, USA

Table 2.4: Continuation

Appliance	Manufacturer
Sartorius precision scale research	Sartorius, Göttingen, Germany
T Professional ThermoCycler (PCR)	Biometra Analytik Jena, Germany
Ultracentrifuge tubes 13.2 ml	Beckman Coutler, Brea, USA
Varioklav steam sterilizer	H+P, Oberschleißheim, Germany
VortexGenie 2	Scientific Industries, Bohemia, USA

2.1.5 Dishes and Pipettes

Table 2.5: Dishes and Pipettes used in this Project

Appliance	Manufacturer
Tissue culture dish, ØxH: 150mmx20mm	Sarstedt, Nümbrecht, Germany
Tissue culture dish, ØxH: 150mmx20mm	Sarstedt, Nümbrecht, Germany
Cell culture plate, 24-well	Sarstedt, Nümbrecht, Germany
Cell culture plate, 96-well	Sarstedt, Nümbrecht, Germany
Nunc Serological Pipette, 2, 5, 10, 25, 50ml	Sarstedt, Nümbrecht, Germany

2.1.6 Cell lines

Table 2.6: Cell lines used in this Project

Cell line	Manufacturer/ATCC number
HEK-293	ATCC CRL-1573TM, Virginia, USA
A549	ATCC CCL-185, Virginia, USA

2.1.7 Software

Table 2.7: Software used in this Project

Software	Supplier
ApE - a Plasmid Editor	Jorgensen Lab
Cytoflex	Beckmann Coulter, Brea, USA
ImageLab	BioRad, Hercules, USA
Overleaf, online $\text{\LaTeX}$ editor	Overleaf c/o Digital Science, London, UK

2.2 Microbial Methods

2.2.1 Bacterial fluid cultures

Only DH10 β E. Coli cells were used in this project. Cells were stored at -80°C and thawed on ice in chilled ddH₂O before use. when growing in fluid culture, LB medium with the appropriate antibiotics was used. The culture temperature was 37 °C, shaking was done at 1,100 rpm if not stated otherwise.

2.2.2 Bacterial culture on agar dishes

To single-out a clonal bacterial population. Lb Agar plates with appropriate antibiotics were prepared and around 100 μ l of bacterial cell suspension, the exact amount varying slightly depending on observed colony density in previous attempts, was streaked on the plate with a sterile serological pipette. Plates were warmed before use and labeled to avoid mixups. the plates were incubated upside down and for a maximum of 16 hours.

2.2.3 Small scale Plasmid DNA preparation

For the small scale preparation (mini-prep) of plasmids, 1.8 ml DH10 β overnight cultures were centrifuged at 16.500 x g and the bacterial pellet re-suspended by vortexing in 100 μ l GTE-Buffer. All of the following centrifuging steps were carried out at 16.500 x g. 200 μ l Lysis-Buffer were added and mixing was done by inverting the tube multiple times, the lysis reaction was neutralized after one minute by adding 150 μ l KAc solution and the tube was inverted a further 4-6 times. Centrifuging was done for 10 minutes at room temperature, the supernatant transferred to a new reaction tube and the DNA precipitated with 1 ml EtoH 100% at -20°C. The centrifuging was repeated for 20 minutes at 4°C. The resulting DNA-pellet was washed with EtoH 70% at -20°C and after a final centrifuging step for 10 minutes at 4°C the pellet was air-dried for five minutes and taken up in 25 μ l TE-Buffer.

Table 2.8: Mini-prep Buffer composition

Component	Amount
GTE-Buffer	25mM Tris pH 8.0, 10mM EDTA pH 8.0, 50mM Glucose, 24 µg/ml RNase A
KaC-Buffer	
Lysis-Buffer (50ml)	1% SDS, 0.2M NaOH

2.2.4 Medium scale Plasmid DNA preparation

When larger quantity of DNA product from recombinant *E. coli* bacteria was needed, growth and extraction in 200 ml format was used. 200 ml of LB medium were supplemented with the appropriate antibiotics to select for the desired recombinant strains and inoculated with a single clonal bacterial culture. The fluid culture was incubated overnight at 37°C while rotating for a maximum of 16 hours. It was then centrifuged for 20 minutes at 5000 x g and 4 °C. Preparation was done with the Genopure Midi prep Kit by Roche and with the manufacturers protocol recommendations. The supernatant was removed and the bacterial pellet suspended in 8ml suspension buffer with added RNase. 8 ml lysis buffer were added and the tube inverted 6-8 times, then incubated 5 minutes at room temperature. 8 ml cold neutralisation buffer were added and incubation continued for another 15 minutes on ice. Afterwards, the lysate was centrifuged for 30 minutes at 18,000 x g at 4°C. Supplied elution columns were treated with 2.5 ml equilibration buffer and the lysate supernatant was given into the column. Flow-through was discarded and the column matrix treated with 4 ml washing buffer 2 times. The column was placed on a fresh falcon and the DNA eluted from the matrix with 5 ml elution buffer. DNA was precipitated with 3.6 ml isopropanol. The additional washing and concentration steps are identical with protocol 2.3.4.

2.2.5 Transformation of DH10 β cells by electroporation

Introduction of foreign DNA into a recombinant *E. Coli* strain was done by electroporation. Cells were thawed, cuvettes and plasmid DNA also chilled on ice and cuvettes were thoroughly dried before use. Immediately after thawing, 5 ng of Plasmid DNA was added to the cell suspension, the mixture

transferred to the cuvette and electroporation was done in at 1,350 V, 10 μ F and 600 Ohms. Optimal electroporation times was aimed to be 5 ms. After the device was finished, the cells were mixed with 1 ml warmed LB medium (37 °C) without antibiotics and shaken for approximately one hour before streaking on agar dishes.

2.3 Molecular biological Methods

2.3.1 DNA storage

Oligonucleotides were stored at -20°C in H₂O. Plasmids and bacmids were stored in TE Buffer with a pH of 8.0 at 4°C.

Table 2.9: TE Buffer composition

Reagents and Amounts
10 mM Tris-HCL, 1 mM EDTA, pH 8.0

2.3.2 Polymerase-Chain-Reaction

PCR is a classic method to produce large amounts of existing genomic material or detect a suspected nucleic acid. For the reaction, DNA primers are needed that originate in sequence from the flanks of the nucleic acids to be amplified. The amplification reaction is catalyzed by a heat-resistant polymerase enzyme. Chronologically, PCR starts with the denaturation of the DNA, then proceeds with the annealing of the primers to the template, and finally with the elongation. For diagnostic use, where the interest was in the size of the resulting PCR products, the GoTaq polymerase was used in combination with the green GoTaq buffer. When preparing material for sequencing, high-fidelity Q5-polymerase with the supplied buffer was used. According to the manufacturers' recommendations, the different polymerases required slightly different reaction conditions, which were applied accordingly. For each reaction, a 50 μ l PCR tube was used. The annealing time was adjusted to the specific primers used in each reaction. Tables 2.10 to 2.13 show the programs and pipetting instructions for each polymerase. Table 2.14 shows the primers used in this project.

Table 2.10: PCR scheme for GoTaq Polymerase

Step	Temperature	Time (min)	Steps
Denaturation	98	2	
Denaturation	98°C	0.5	x35
Annealing	X	0.75	
Elongation	72°C	0.5/kb	
Elongation	72°C	15	

Table 2.11: PCR scheme for Q5 Polymerase

Step	Temperature	Time (min)	Steps
Denaturation	98	2	
Denaturation	98°C	0.5	x35
Annealing	X	0.75	
Elongation	72°C	1/kb	
Elongation	72°C	20	

Table 2.12: basic pipetting scheme for GoTaq polymerase

Component	Volume
x5 green GoTaq Buffer	10 μ l
dNTPs	1 μ l
Forward primer (10 μ M)	1 μ l
Reverse primer (10 μ M)	1 μ l
Template DNA	50 ng
GoTaq polymerase	0.25 μ l
H ₂ O	36,25 μ l

Table 2.13: basic pipetting scheme for Q5 polymerase

Component	Volume
x5 Q5 Buffer	10 μ l
dNTPs	1 μ l
Forward primer (10 μ M)	2.5 μ l

Table 2.13: Continuation

Component	Volume
Reverse primer (10 μM)	2.5 μl
Template DNA	300 ng
Q5 polymerase	0.5 μl
H ₂ O	33 μl

Table 2.14: Primers used in this project

Name	Sequence
Left junction fwd 1 (711)	GAACCAATATGAACTGCAGGC
Left junction fwd 2 (713)	CTAAAAGAGCGCCTACATGG
Right junction rev 1 (716)	GAGGATTCTGTTCAGAGACCTG
Right junction rev 2 (718)	AAATGGTGAGAGCCATCCAC
E1a fwd (763)	TACCCGCCGTCCTAAAATGG
E1b rev (766)	GGAACAGCGGGTCAGTATGT

2.3.3 Agarose Gel Electrophoresis

DNA Gel electrophoresis utilises the proportionally increasing, length-dependent travel speed of negatively charged DNA molecules through a gel matrix with applied electric current. Travel over time can be influenced by varying the pore size of the gel matrix or the voltage applied to the gel. In this work, 0.8% agarose gels were used. The gels were prepared with the addition of one drop ethidium bromide, a fluorescent DNA intercalating agent, for identifying fragment sizes. The runtime varied from 30 to 50 minutes at a voltage of 5 V/cm, varying slightly depending on expected fragment sizes. A Gel Doc EZ camera system was used to document the finished gels.

2.3.4 Elution of DNA from agarose gels

When further use of DNA fragments after electrophoresis (2.3.3) was intended, the position of the desired band was confirmed by inspection under UV-light and the gel was cut out with a scalpel and minced finely. a 1:1 ratio by weight of phenol was added, the reaction tube shaken vigorously for one minute and frozen in liquid nitrogen for an additional minute before being thawed completely in hand. The tube was centrifuged for 15 minutes at 19,000 x g (same g s for all centrifuging steps in this protocol). the upper aqueous phase was separated into a new reaction tube. a 1:1 ratio of chloroform-isoamylalcohol 24:1 was added, shaking and spinning were repeated like above. the upper phase was again separated and 0.1 volumes 3M NaAc pH5.2 along with 0.7 volumes isopropanol 100% were added. This was left in the freezer at -20°C for 30 minutes or alternatively overnight. Centrifuging was repeated and the resulting pellet solved in 150 μ l TE buffer. To further enhance DNA purity, precipitation was repeated with 0.1 volumes NaAc and 2.5 volumes Ethanol 100%. The Centrifuged pellet was then washed with Ethanol 70%, centrifuged and solved in 30 μ l TE buffer.

2.3.5 px330 expression plasmid and BbsI restriction enzyme

The plasmid px330, an aliquot of which was kindly given to us by Dr. Eric Schulz from UWH, carries the Cas9-sequence, barring the sequence of the guide-rna, and a puromycin resistance encoding sequence.

BbsI is a restriction enzyme originating from *Bacillus brevis*, recognising a single six-nucleotide motif. The px330 plasmid has two matching bbsI-sites, that are cut away after bbsI-restriction and insertion of a gRNA-sequence.

2.3.6 DNA oligo preparation

Single-stranded DNA oligos with the gRNA-sequences were ordered in complimentary forward and reverse orientation from ThermoFisher. The stocks were diluted to a concentration of 1 $\mu\text{g}/\mu\text{l}$. 1 μg each of matching forward and reverse oligos were pipetted ad 50 μl TE 8.0 in a reaction tube and boiled in water on a heating plate for 15 minutes. the heating plate was then turned off and the oligos were left to cool overnight at room temperature. Table 2.15 shows the sequences of the gRNA oligos

Table 2.15: gRNA oligo sequences

Name	Sequence
gRNA 1 fwd	CACCGGGCGGAACACATGTAAGCGA
gRNA 1 rev	AAACTCGCTTACATGTGTTCCGCCC
gRNA 2 fwd	CACCGGTAAACGGTCAAAGTCCCCG
gRNA 2 rev	AACCGGGGACTTTGACCGTTTACC
gRNA 3 fwd	CACCCCGTCGGTATTATCGCCGGG
gRNA 3 rev	AAACCCCGGCGATAATACCGACGGC
gRNA 4 fwd	CACCTGTACAACATTTCATCCCGA
gRNA 4 rev	AAACCGTTCAGGTTACCTTGGACC
gRNA 5 fwd	CACCGTCCAAGGTGAACCTGAACG
gRNA 5 rev	AAACCGTTCAGGTTACCTTGGACC
gRNA 6 fwd	CACCGTGTTTACCGCCACACTCGCA
gRNA 6 rev	AAACTGCGAGTGTGGCGGTAAACAC

2.3.7 Restriction-ligation cloning of DNA oligos into plasmid backbones

To linearize the 10 ng of px330 plasmid was incubated with 20 units BbsI ad 100 μl H₂O supplemented with appropriate measure of NEB 2.1 enzyme buffer at 37 °C for one hour. Then, the whole reaction solution was subjected to gel electrophoresis like described under 2.3.3 to separate successfully linearized plasmid and purified from the finished gel run like shown in 2.3.4.

The plasmid DNA concentration of the eluate was then measured and amounts for the ligation mixed like in table 2.16. Table

Table 2.16: ligation reaction pipetting scheme

Component	Amount
10x T4 ligase Buffer	1 μ l
lin. px330 plasmid	100 ng
ds gRNA oligo	0.55 μ l
T4 ligase	0.5 μ l
H ₂ O	5.95 μ l

First, only H₂O and the plasmid were mixed, incubated for 10 minutes at 42°C to linearize any supercoiled DNA structures that would impede enzyme efficacy. The mixture was then cooled for 5 minutes on ice. Buffer, ligase and oligos were added to the reaction tubes and the ligation was catalyzed at 16°C overnight in a thermocycler.

2.4 Cell biological Methods

2.4.1 Eucaryotic Cell culture

A549 and HEK293 cells were used. They were cultured in 15 cm dishes and kept in MEM medium supplemented with 10% FBS and 1% penicillin / streptomycin. HEK cells were split at around 80-90 percent confluency. To split the cells, they were washed with PBS, treated with Trypsin for 1 minute before stopping the reaction with unsupplemented MEM medium and then repeatedly pipetting up and down. Depending on the confluency, the splitting ratio was 1:8 to 1:10. A549 cells were treated in the same manner, except cells were trypsinised for 3 minutes and split at a ratio of 1:6 to 1:8. All reagents used were warmed to 37 °C before being used. Dishes were stored at 37 °C in an incubator with an atmosphere of 5% CO₂.

2.4.2 Cell Counting

To determine cell numbers for usages where exact quantities are needed, cells were removed from their dishes and separated like described above and resuspended in MEM medium. An improved Neubauer chamber was loaded with the cell suspension and all four squares were counted under the light-microscope. the mean of those squares was then taken to calculate cell count like

$$[Cells/ml] = [Mean/squares]1 * 10^4$$

This value was then used to calculate the desired volume to use in cell seeding.

2.4.3 Transfection of HEK293 cells with Cas9 Plasmids

24 hours prior to transfection HEK293 cells were seeded in 6cm cell culture dishes counted to an amount of 5×10^5 per dish. Dishes were filled with 2 ml of medium. All gRNA constructs were transfected in parallel. Per gRNA, 500 ng were used, 3 μ g for each cell culture dish in total. Transfection was prepared by mixing the correct amounts of plasmid solution and filling up with 150 mM NaCl to a volume of 250 μ l per transfection aliquot. The appropriate measure of Polyethylenimine (PEI) was calculated with the following formula:

$$PEI [\mu l] = (\mu g DNA) \times 12$$

As calculated with the used DNA amounts, 36 μ l PEI were diluted to a volume of 250 μ l 150 mM NaCl for each transfection. PEI is a molecule that forms complexes with DNA of positive net charge, allowing for higher rates of intracellular DNA delivery that would normally be hindered by the opposing negative charges of the DNA and the cellular membrane. The NaCl solution containing the PEI was given into the DNA aliquots carefully, additionally pipetted up and down three times and then incubated at room temperature for 10 minutes. Finally the mixture was added to the cell dishes after they were washed with PBS and supplemented with fresh Medium. After at least 2 hours in the incubator, medium was filled up to a volume of 6 ml per dish.

2.4.4 Extraction of genomic DNA

To yield high purity genomic DNA, to use for sequencing or PCR-assays, the phenol-chloroform extraction method was used in this project. Cells were removed from 15 cm culture dishes as described in 2.4.1. for one 1.5 ml microcentrifuge tube, half of the cells from a confluent 15 cm dish were used. Medium was removed by centrifuging at 300 x g for 4 minutes and the cell pellet was re-suspended in 200 μ l PBS. 200 μ l Lysis-Buffer was added and the suspension was mixed thoroughly before adding 30 μ l SDS 10% and 20 μ l proteinase K. The mixture was shaken overnight at 55°C and 1,200 rpm. After overnight shaking, approximately 16 hours later, 20 μ l RNase A was added and the solution was incubated for a further 30 minutes at 37°C/900rpm. 470 μ l Phenol was added and the tube shaken by hand for one minute, then centrifuged at 11,000 x g for 10 minutes and resulting supernatant was carefully transferred into another microcentrifuge tube and 450 μ l Phenol was added. Shaking, centrifuging and transferring were repeated twice as above, swapping phenol for 430 μ l chloroform/isoamylalcohol in a ratio of 24:1 before the second repetition. 40 μ l NaAc and 1ml Ethanol 100% were added and the tube was inverted until precipitated DNA could be visually confirmed in the tube. centrifuging at full speed for 20 minutes at room temperature yielded a DNA pellet which was separated from the supernatant and washed with 500 μ l Ethanol 50%, centrifuged at full speed for 5 minutes, air-dried for one minute and suspended in 50 μ l TE-Buffer before being stored at 4°C overnight to re-enter solution. DNA quantity in the solution was then measured like described in 2.4.5.

Table 2.17: Lysis buffer composition

Reagents and Amounts
10 mM Tris ultrapure, 10 mM EDTA, 0,5 % SDS

2.4.5 genomic DNA quantification

To approximate the DNA concentration in a given solution, OD_{260} was measured using a photometer, which was calibrated using a blank solution. DNA concentration was then calculated by the device and displayed as ng/ μ l.

Table 2.18: Composition of solutions for DNA quantification

Solution	Composition
blank solution	1 μ l TE 8.0, 69 μ l H ₂ O
measured solution	1 μ l DNA elution, 69 μ l H ₂ O

2.4.6 Whole-cell protein extraction

To prepare protein extract for SDS-page, cells were seeded in 12 well cell culture dishes and grown to $\simeq$ 60% confluency. Cells were then washed with PBS and treated with 300 μ l lysis buffer for 10 minutes at room temperature. 100 μ l TCA 100% was then added and mixing was done by gently shaking the culture dish. The cloudy supernatant was transferred to a 1.5 ml reaction tube, leaving a small white sphere of denatured DNA in the dish well. the tube was centrifuged for 5 minutes at 5,000 x g, room temperature and the pellet washed with TCA 2.5%. Centrifuging was repeated like above and the pellet washed with 20 μ l tris-base and vortexed briefly. The tube was incubated for 30 minutes at room temperature and then suspended in 20 μ l H₂O to re-enter solution. Protein extract was stored at -20°C in the freezer.

2.4.7 SDS-Page

Table 2.19: SDS-Page Buffers and Solutions

Buffer	Composition
10% Separating Gel	6 ml dH ₂ O, 3.75 ml 4x Separating Gel Buffer pH 8.8, 5 ml 30% PAA (29:1), Combine and use 5 ml of the solution, 50 µl 10% APS, 5 µl TEMED
5% Stacking Gel	3.4 ml dH ₂ O, 1.5 ml 4x Stacking Gel Buffer pH 6.8. 1 ml 30% PAA (29:1), Combine and use 2.5 ml of the solution, 30 µl 10% APS, 3 µl TEMED
Seperating Gel Buffer	
Stacking Gel Buffer	

The separation gel was first poured into the a gel preparation chamber. Before pouring the stacking gel, a few milliliters of isopropanol were added to remove any bubbles. after 10 minutes, the isopropanol was removed and the stacking gel poured. Protein samples were mixed with SDS loading buffer and denatured for 10 minutes at 75°C. Voltage was set to 4V/cm and increased to 5 V/cm once the dye front reached the Separating gel. The gel-chamber was placed on ice for the whole run to protect the gel matrix integrity.

2.4.8 Western Blot

Table 2.20: Western Blot Buffer composition

Buffer	Composition
10x Tris-Glycin buffer	30 g/l Tris pH 8,3, 144 g/l Glycin
1x Transfer buffer (1L)	100 ml 10x Tris-Glycin, 200 ml Methanol, 700 ml H ₂ O
10x TBS (1L)	100 mM Tris, 1.5 M NaCl, 60 ml HCl ad 1L ddH ₂ O ph 7.4
1x TBS-T	450 ml H ₂ O, 50 ml 10x TBS, 250 µl Tween-20

Table 2.20: Continuation

Buffer	Composition
Blocking buffer	5% BSA in TBS-T
Primary antibody	1:1,000 in blocking buffer
Secondary antibody	1:15,000 in blocking buffer

Gels resulting from SDS-PAGE (2.4.7) were subjected to lateral electric current of 5 V/cm for one hour in an ice-cooled blotting chamber. The gel was placed adjacent to a nitrocellulose membrane, thus allowing travel of the proteins along the current onto the secondary membrane. After transfer, the membrane was washed in TBS, shortly air-dried and placed in blocking buffer overnight at 4°C. After blocking, the membrane was incubated with the primary antibody for 1 hour at room temperature and then rinsed 5 times in TBS-T. After rinsing, incubation with the secondary antibody was performed for 45 minutes at room temperature and rinsing was repeated as done above. The membrane was then briefly dried and signals were detected in the Odyssey CLx Infrared Scanner.

2.4.9 Nuclear staining of E1a

Table 2.21: Nuclear stain Buffer composition

Buffer	Contents
Fixation buffer	4% PFA in PBS
Permeabilization buffer	0.1% Triton X-100, 0.68 μ M EDTA, 1% BSA in PBS
Incubation buffer	0.5% BSA in PBS

Cells were prepared for seeding like described in 2.4.2. 5×10^5 cells were used per well of a conical 96-well plate. after detatching, cells were centrifuged and re-suspended in 100 μ l fixation buffer at room temperature for 15 minutes, then centrifuged again. All centrifuging steps, unless otherwise stated, are performed at 4°C, 300 x g, for 5 Minutes. the cells were mixed with 50 μ l permeabilization buffer and incubated on ice, 10 minutes, then centrifuged, washed in 100 μ l incubation buffer, centrifuged and 50 μ l incubation buffer with the diluted first antibody was added. The incubation time for the first antibody was 45 minutes at 4 °C, re-suspending

the cells once to ensure even distribution in the solution. After incubation, cells were spun, immediately followed by mixing with 50 μ l incubation buffer with the diluted second antibody. incubation times and temperature was repeated like before and finally, cells were washed with incubation buffer and re-suspended in 150 μ l of incubation buffer and transferred to a 96 well FACS plate and proceeding with the measurements.

2.4.10 Ad-vector replication Assay

To indirectly assess E1-expression, E1-knockdown HEK293 clones were infected with a recombinant Ad5 vector expressing eGFP under control of the MLP promoter, which is active only when replication occurs. Since replication of E1-deficient Ad5 vectors is dependent on cellular E1 transcomplementation, changes in E1 expression can be extrapolated by measuring eGFP fluorescence with flow cytometry. Cells were counted to 2×10^4 /100 μ l medium and seeded in 96-well cell culture dishes. 24 hours later, cells were infected with a MOI of 300 in a volume of 20 μ l cell culture medium. 28 hours post infection. cells were fixated and prepared and measured like described under 2.4.11

2.4.11 Flow Cytometry with Eucaryotic Cells

In this project, methods 2.4.9 and 2.4.10 used flow cytometry (FC) measurements to generate Data. Prior to FC, cells had the medium removed, were removed from dish binding with trypsin, after 1 minute the reaction was halted with medium. The cells were spun for 5 minutes at 400 x g, room temperature, the pellet re-suspended in PFA 4% and incubated for 15 minutes. Centrifuging was repeated and the cells suspended in 130 μ l PBS and transferred to a FC plate. In FC settings, gating for single cells was done for all measurements and data was extracted as .csv.

2.4.12 Statistical analysis

2.4.13 DNA Sequencing

DNA sequencing was done using sanger-sequencing by the Microsynth AG, DNA was prepared as per the requirements of the service provider: For plasmid DNA the preparation was 50 ng/ μ l in a total Volume of 12 μ l

H₂O. For PCR products, varying concentrations depending on the fragment length were used.

Chapter 3

Results

- 3.1 junctions of Ad5 E1 and genomic DNA were amplified and sequenced**
- 3.2 Six custom gRNA sequences were successfully cloned into the Cas9 plasmid backbone**
- 3.3 HEK293-Cells were transfected, selected and genomic DNA was extracted**
- 3.4 PCR-screening of gDNA revealed altered amplicons in several cell populations**
- 3.5 Sanger sequencing confirmed four of six desired DSB-locations**

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Appendix A

Appendix

List of Figures

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Table A.1: List of Abbreviations

Abbreviation	Meaning
Ad5	Adenovirus type 5
crRNA	CRISPR-RNA
sgRNA	single guide RNA

Sequence of the Cas9 expression plasmid

```

1 CGTAGAAAA ATCAAGGAT CTCTTGAGA TCTTTTTT CTGCGGTAA TCTGCTGCTT GCAACAAAA AAACCACCG TACCAGCGGT GGTGTTGTTG
101 CCGGATCAAG AGCTACCAAC TCTTTTTCCG AAGGTAAC TGCTTCAGCAG AGCGCAGATA CCAAATCTG TCCTTCTAGT GTAGCCGTAG TTAGGCCACC
201 ACTTCAAGAA CTCTGTAGCA CCGCTACAT ACCTCGCTCT GCTAATCCTG TTACCACTGG CTGCTGCCAG TGGCGATAAG TCGTGTCTTA CCGGGTTGGA
301 CTCGAAGAGA TAGTTACCGG ATAAGGCCA GCGGTCGGGC TGAACGGGGG GTTCGTGCAC ACAGCCGAGC TTGGAGCGAA CGACCTACAC CGAACTGAGA
401 TACCTACAGC GTGAGCTATG AGAAAGCGCC ACGCTTCCCG AAGGGAGAAA GGCGGACAGG TATCGGTAA GCGGCGAGGT CGGAACAGGA GAGCGCACGA
501 GGGAGCTTCC AGGGGAAAC GCCTGGTATC TTTATAGTCC TGTCGGGTTT CGCCACCTCT GACTTGAGCG TCGATTTTTG TGAATGCTGT CAGGGGGGCG
601 GAGGCTATGG AAAACCGCCA GCAACGCGGC CTTTTTACGG TTCTGGGCTT TTGCTGGCC TTTTGCTCAC ATGTGAGGGC CTATTTCCCA TGATTCCTTC
701 ATATTTGCTG ATACGATACA AGGCTGTTAG AGAGATAATT GGAATTAATT TGACTGTAAA CACAAAGATA TTAGTACAAA ATACGTGACG TAGAAAGTAA
801 TAATTTCTTG GTGAGTTTGC AGTTTTAAAA TTATGTTTTA AAATGGAATA TCATATGCTT ACCGTAACTT GAAAGTATTT CGATTTCTTG GCTTTATATA
901 TCTTGTGAAA AGGACGAAAC ACCGGGTCTT CGAGAAGACC TGTTTTAGAG CTAGAAATAG CAAGTTAAAA TAAGGCTAGT CCGTTATCAA CTTGAAAAAG
1001 TGGCACCAGC TCGGTGCTTT TTTGTTTTAG AGCTAGAAAT AGCAAGTTAA AATAAGGCTA GTCCGTTTTT AGCGCGTGC GCAATTCTGC AGACAAATGG
1101 CTCTAGAGGT ACCCGTTACA TAACTTACGG TAAATGGCCC GCCTGGCTGA CCGCCCAACG ACCCCGCGCC ATTGACGTCA ATAGTAACGC CAATAGGAC
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8401 CCGGTATTGA CGCCGGGCAA GAGCAACTCG GTGCGCGCAT ACACTATTCT CAGAAATGACT TGGTTGAGTA CTCACCACTC ACAGAAAAGC ATCTTACGGA
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8601 ACCGCTTTTT TGCACAACAT GGGGATCAT GTAACCTGCG TTGATCGTTG GGAACCGGAG CTGAATGAAG CCATACCAA CGACGAGCT GACACCACGA
8701 TGCGGTAGC AATGGACAAC ACGTTGCGCA AACTATTAA TGGCGAACTA CTTACTTAG CTTCGCGCA ACAATTAATA GACTGGATGG AGCGGATAA
8801 AGTTGCAGGA CCACTTCTGC GCTCGGCCCT TCCGCTGGC TGGTTTATTG CTGATAAATC TGGAGCGGCT GAGCGTGGAA CCGCGGTGAT CATTGCAGCA
8901 CTGGGGCCAG ATGGTAAGC CTCCGTATC GTAGTTATCT ACACGACGGG GAGTCAGGCA ACTATGGATG AACGAAATAG ACAGATCGCT GAGATAGGTG
9001 CCTCACTGAT TAAGCATTGG TAACTGTGAG ACCAAGTTTA CTCATATATA CTTTAGATTG ATTTAAAACT TCATTTTTAA TTTAAAAGGA TCTAGGTGAA
9101 GATCCTTTTT GATAATCTCA TGACCAAAAT CCCTTAACGT GAGTTTTCGT TCCACTGAGC GTCAGACC